

Proof: Suppose $\dim W = m$. Let $B_W = \{\vec{w}_1, \dots, \vec{w}_m\}$ be a basis of W . Let $A = [T|_W]_{B_W} \in \mathbb{C}^{m \times m}$

Denote $A = (a_{ij})$. Then $T\vec{w}_j = \sum_{i=1}^m a_{ij} \vec{w}_i \Rightarrow f_W(x) = \det(A - \lambda I)$

Extend B_W to a basis of V : $B = \{\vec{w}_1, \dots, \vec{w}_m, \vec{w}_{m+1}, \dots, \vec{w}_n\}$.

$$\text{Then } T\vec{w}_j = \sum_{i=1}^m a_{ij} \vec{w}_i, \quad j=1, \dots, m \quad T\vec{w}_k = \sum_{i=1}^n a_{ik} \vec{w}_i$$

$$\Rightarrow [T]_B = \left(\begin{array}{c|c} A & M \\ \hline 0 & C \end{array} \right)$$

$$\Rightarrow f(x) = \det([T]_B - \lambda I) = \det(A - \lambda I) \det(C - \lambda I) = f_W(x) \det(C - \lambda I)$$

$$\Rightarrow f_W(x) \mid f(x).$$

Proposition: Let $T: V \rightarrow V$ be a linear transformation. Let λ be an eigenvalue of T with algebraic multiplicity λ .

1). K_λ is T -invariant

2). K_λ is $(T - \lambda I)$ -invariant

3). K_λ is $(T - \mu I)$ -invariant, $\forall \mu \in \mathbb{C}$

4). If $\mu \neq \lambda$, then by 4). we can define: $(T - \mu I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$. Then $(T - \mu I)|_{K_\lambda}$ is one-to-one.

5). If $\mu \neq \lambda$ is another eigenvalue of T , then $K_\lambda \cap K_\mu = \{\vec{0}\}$.

6). $\dim(K_\lambda) \leq m_\lambda$

7). $K_\lambda = \text{Nul}(T - \lambda)^{m_\lambda}$.

Proof: $\forall \vec{x} \in K_\lambda$. Then $\exists p \in \mathbb{Z}^+$ s.t. $(T - \lambda I)^p(\vec{x}) = \vec{0}$.

$$\text{Then } 1) (T - \lambda I)^p(T(\vec{x})) = T((T - \lambda I)^p(\vec{x})) = T(\vec{0}) = \vec{0}$$

$$\Rightarrow T(\vec{x}) \in K_\lambda. \quad \Rightarrow K_\lambda \text{ is } T\text{-invariant.}$$

$$2) (T - \lambda I)^p((T - \lambda I)(\vec{x})) = (T - \lambda I)^{p+1}(\vec{x}) = \vec{0}$$

$$\Rightarrow (T - \lambda I)(\vec{x}) \in K_\lambda \quad \Rightarrow K_\lambda \text{ is } (T - \lambda I)\text{-invariant.}$$

$$3). (T-\lambda I)^p (T-\mu I)\vec{x} = (T-\mu I) (T-\lambda I)^p (\vec{x}) = \vec{0}$$

$$\Rightarrow (T-\mu I)(\vec{x}) \in K_\lambda \Rightarrow K_\lambda \text{ is } (T-\mu I)\text{-invariant.}$$

4). As K_λ is $(T-\mu I)$ -invariant, we consider a restriction map: $T-\mu I|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$.

Suppose, towards a contradiction, $\text{Null}((T-\mu I)|_{K_\lambda}) \neq \{\vec{0}\} \Rightarrow$ let $\vec{x} \in \text{Null}((T-\mu I)|_{K_\lambda})$ such that $\vec{x} \neq \vec{0}$.

$$(T-\mu I)(\vec{x}) = \vec{0}. \text{ As } \vec{x} \in K_\lambda \exists \text{ smallest } p \in \mathbb{Z}^+, \text{ s.t. } (T-\lambda I)^p(\vec{x}) = \vec{0}.$$

$$\text{By the choice of } p: \vec{y} = (T-\lambda I)^{p-1}(\vec{x}) \neq \vec{0}.$$

$$(T-\lambda I)(\vec{y}) = (T-\lambda I)(T-\lambda I)^{p-1}(\vec{x}) = \vec{0} \Rightarrow \vec{y} \in E_\lambda.$$

$$(T-\mu I)(\vec{y}) = (T-\mu I)(T-\lambda I)^{p-1}(\vec{x}) = (T-\lambda I)^{p-1}(T-\mu I)(\vec{x}) = \vec{0} \in \vec{y} \in E_\mu.$$

$$\text{As } \lambda \neq \mu: E_\lambda \cap E_\mu = \{\vec{0}\}. \Rightarrow \vec{y} = \vec{0} \text{ contradiction.}$$

5). Suppose $\vec{x} \in K_\lambda \cap K_\mu$. $\exists p, q \in \mathbb{Z}^+$ such that $(T-\lambda I)^p(\vec{x}) = \vec{0}$ and $(T-\mu I)^q(\vec{x}) = \vec{0}$.

As $(T-\mu I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$ is one-to-one, $(T-\mu I)^q|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$ is also one-to-one

$$\text{As } \vec{x} \in K_\lambda, (T-\mu I)^q(\vec{x}) = \vec{0} \Rightarrow \vec{x} = \vec{0}.$$

6). As m_λ is the multiplicity of λ , which is an eigenvalue of $T: V \rightarrow V$.

Then the characteristic polynomial of T has the form: $f(x) = (\lambda - x)^{m_\lambda} g(x)$ where

$$(\lambda - x) \nmid g(x). \text{ i.e. } g(x) \text{ doesn't have the factor } (\lambda - x).$$

Assume $\dim(K_\lambda) = m$. Since K_λ is T -invariant, we may consider the restriction map:

$$T|_{K_\lambda}: K_\lambda \rightarrow K_\lambda. \text{ let } f_{K_\lambda}(x) \text{ be the characteristic polynomial of } T|_{K_\lambda}.$$

$$\text{then } f_{K_\lambda}(x) \mid f(x).$$

Claim: λ is the only eigenvalue of $T|_{K_\lambda}$ on K_λ . end of Month 7

Proof of the claim: ① λ is an eigenvalue of $T|_{K_\lambda}$ on K_λ . In fact $E_\lambda \subseteq K_\lambda$

$$\Rightarrow \exists \vec{x} \neq \vec{0} \text{ and } \vec{x} \in E_\lambda \subseteq K_\lambda \text{ such that } (T-\lambda I)\vec{x} = \vec{0} \Leftrightarrow (T|_{K_\lambda} - \lambda I)\vec{x} = \vec{0}$$

$$\Rightarrow \vec{x} \text{ is an eigenvector of } T|_{K_\lambda} \text{ on } K_\lambda \text{ associated with eigenvalue } \lambda.$$

② For any $\mu \neq \lambda$, μ is not an eigenvalue of $T|_{K_\lambda}$ on K_λ . In fact
 if $\exists \vec{x} \in K_\lambda$ such that $(T - \mu I)(\vec{x}) = \vec{0} \Leftrightarrow (T|_{K_\lambda} - \mu I)(\vec{x}) = \vec{0}$
 Since $(T - \mu I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda$ is one-to-one $\vec{x} = \vec{0}$

Followed by the claim: $f_{K_\lambda}(x) = (\lambda - x)^m$. Then $(\lambda - x)^m \mid (\lambda - x)^{m_\lambda} g(x)$.

As $(\lambda - x)$ is not a factor of $g(x)$, $m \leq m_\lambda$. i.e. $\dim(K_\lambda) \leq m_\lambda$.

7). Use the notation in 6). by Cayley-Hamilton: $\forall \vec{x} \in K_\lambda: f_{K_\lambda}(T|_{K_\lambda})(\vec{x}) = \vec{0}$
 $\Leftrightarrow (\lambda I - T|_{K_\lambda})^m(\vec{x}) = \vec{0}$ for any $\vec{x} \in K_\lambda$
 $\Leftrightarrow (\lambda I - T)^m(\vec{x}) = \vec{0}$ for any $\vec{x} \in K_\lambda$
 $\Rightarrow \vec{x} \in \text{Null}(\lambda I - T)^m \subseteq \text{Null}(\lambda I - T)^{m_\lambda}$
 $\Rightarrow K_\lambda \subseteq \text{Null}(\lambda I - T)^{m_\lambda}. \quad (*)$

By definition: $K_\lambda = \{ \vec{x} : (T - \lambda I)^p(\vec{x}) = \vec{0} \text{ for some } p \in \mathbb{Z}^+ \}$.

$\Rightarrow \text{Null}(\lambda I - T)^{m_\lambda} \subseteq K_\lambda \quad (**)$

By (*) and (**): $K_\lambda = \text{Null}(T - \lambda I)^{m_\lambda}. \quad \square$

Proposition: Let $T: V \rightarrow V$ be a linear transformation. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be all the distinct eigenvalues of

T with multiplicities $m_1, m_2, \dots, m_k$. Then $\forall \vec{x} \in V: \exists \vec{x}_1 \in K_{\lambda_1}, \dots, \vec{x}_k \in K_{\lambda_k}$ such that

$$\vec{x} = \vec{x}_1 + \vec{x}_2 + \dots + \vec{x}_k$$

$k=2$ case:

Let $W_1 = V$

$W_2 = \text{Range}((T - \lambda_1 I)^{m_1}|_{W_1}) \Rightarrow W_2$ is T -invariant.

Then $T|_{W_2}$ has only λ_2 as its eigenvalue. And $(T - \lambda_2 I)^{m_2}(K_{\lambda_2} \cap W_2) = K_{\lambda_2} \cap W_2$

$$\Rightarrow W_2 \subseteq K_{\lambda_2}.$$

$$\forall \vec{x} \in V \quad (T - \lambda_1 I)^{m_1}(\vec{x}) \in W_2$$

$$\Rightarrow \exists \vec{y}_2 \in K_{\lambda_2} \cap W_2 : \quad \text{s.t. } (T - \lambda_1 I)^{m_1}(\vec{x}) = \vec{y}_2$$

$$\text{Since } \vec{y}_2 \in W_2, \exists \vec{x}_2 \in V \text{ s.t. } \vec{y}_2 = (T - \lambda_1 I)^{m_1}(\vec{x}_2)$$

$$\Rightarrow (T - \lambda_1 I)^{m_1}(\vec{x}) = (T - \lambda_1 I)^{m_1}(\vec{x}_2)$$

$$\Rightarrow (T - \lambda_1 I)^{m_1}(\vec{x} - \vec{x}_2) = \vec{0}$$

$$\Rightarrow \vec{x} - \vec{x}_2 \in \text{Null}(T - \lambda_1 I)^{m_1} = K_{\lambda_1}$$

$$\Rightarrow \exists \vec{x}_1 \in K_{\lambda_1}, \text{ such that } \vec{x} - \vec{x}_2 = \vec{x}_1$$

$$\Rightarrow \vec{x} = \vec{x}_1 + \vec{x}_2.$$

As another example, let's look at the case when $n=3$. i.e. $T: V \rightarrow V$ has distinct eigenvalues $\lambda_1, \lambda_2, \lambda_3$

with multiplicities m_1, m_2, m_3 , respectively. *... took away K_{λ_1}*

$$\text{Then: } W_1 = \text{Range}((T - \lambda_1 I)^{m_1} | V) \Rightarrow T|_{W_1} \text{ has only eigenvalues of } \lambda_2, \lambda_3, \quad (T - \lambda_1 I)^{m_1}(K_2) = K_2$$

$$W_2 = \text{Range}((T - \lambda_2 I)^{m_2} | W_1) \Rightarrow T|_{W_2} \text{ has only eigenvalues of } \lambda_3 \text{ and } (T - \lambda_2 I)^{m_2}(K_3) = K_3$$

$$W_3 = \text{Range}((T - \lambda_3 I)^{m_3} | W_2) = \{\vec{0}\}.$$

$$\forall \vec{x} \in V: (T - \lambda_2 I)^{m_2} (T - \lambda_1 I)^{m_1}(\vec{x}) \in W_3.$$

$$\exists \vec{z}_3 \in K_{\lambda_3} \cap W_2 \quad (T - \lambda_2 I)^{m_2} ((T - \lambda_1 I)^{m_1}(\vec{x})) = \vec{z}_3$$

$$\exists \vec{y}_3 \in W_1 \cap K_{\lambda_3} \quad \text{s.t. } \vec{z}_3 = (T - \lambda_2 I)^{m_2}(\vec{y}_3).$$

$$(T - \lambda_2 I)^{m_2} ((T - \lambda_1 I)^{m_1}(\vec{x}) - \vec{y}_3) = \vec{0}$$

$$\Rightarrow \underbrace{(T - \lambda_1 I)^{m_1}(\vec{x})}_{\in W_1} - \underbrace{\vec{y}_3}_{\in W_1} \in K_{\lambda_2} \cap W_1 \Rightarrow \exists \vec{y}_2 \in K_{\lambda_2} \cap W_1, \quad \text{s.t. } (T - \lambda_1 I)^{m_1}(\vec{x}) - \vec{y}_3 = \vec{y}_2$$

$$\text{As } y_2, y_3 \in W_1: \exists \vec{x}_2, \vec{x}_3 \in V \text{ s.t. } \vec{y}_2 = (T - \lambda_1 I)^{m_1} \vec{x}_2$$

$$\vec{y}_3 = (T - \lambda_1 I)^{m_1} \vec{x}_3.$$

$$\Rightarrow (T - \lambda_1 I)^{m_1} (\vec{x} - \vec{x}_3 - \vec{x}_2) = \vec{0}$$

$$\Rightarrow \vec{x} - \vec{x}_3 - \vec{x}_2 \in K_{\lambda_1} \cap V$$

$$\Rightarrow \exists \vec{x}_1 \in K_{\lambda_1} \cap V \quad \text{and} \quad \vec{x} - \vec{x}_3 - \vec{x}_2 = \vec{x}_1 \Rightarrow \vec{x} = \vec{x}_1 + \vec{x}_2 + \vec{x}_3. \quad \square$$

Proof: Induction on the number of distinct eigenvalues of T on T -invariant subspaces.

Base Case: If $W \subseteq V$ is a T -invariant subspace and $T|_W$ has exactly one eigenvalue λ . Then

$$\forall \vec{x} \in W: \vec{x} \in K_{\lambda} \cap W.$$

Inductive hypothesis:

Assume $W \subseteq V$ is a T -invariant subspace and $T|_W$ has $s-1$ ^{distinct} eigenvalues of

$$\lambda_1, \dots, \lambda_{s-1}. \quad \text{and} \quad \forall \vec{x} \in W: \exists \vec{x}_1 \in K_{\lambda_1} \cap W, \dots, \vec{x}_{s-1} \in K_{\lambda_{s-1}} \cap W.$$

$$\text{s.t. } \vec{x} = \vec{x}_1 + \vec{x}_2 + \dots + \vec{x}_{s-1}.$$

Inductive steps

let $Z \subseteq V$ be a T -invariant subspace and $T|_Z$ has s ^{distinct} eigenvalues of

$$\lambda_1, \lambda_2, \dots, \lambda_{s+1}. \quad \text{We need show } \forall \vec{x} \in Z: \exists \vec{x}_1 \in K_{\lambda_1} \cap Z, \dots, \vec{x}_s \in K_{\lambda_s} \cap Z$$

$$\text{s.t. } \vec{x} = \vec{x}_1 + \dots + \vec{x}_s.$$

Proof of "base case." let W be an T -invariant subspace, and λ is the only eigenvalue of $T|_W$. Then the characteristic polynomial of $T|_W: W \rightarrow W$

is of the form $f(x) = (\lambda - x)^m$ for some $m = \dim W$.

$$\text{By Cayley-Hamilton Theorem. } f(T|_W) = (\lambda I - T|_W)^m = 0$$

$$\Rightarrow \forall \vec{x} \in W: (\lambda I - T|_W)^m (\vec{x}) = \vec{0} \Rightarrow \vec{x} \in K_{\lambda} \cap W.$$

Proof of inductive steps: Suppose $T|_Z: Z \rightarrow Z$ has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_s$ with multiplicities $m_1, m_2, \dots, m_s$, respectively.

Note for each $i=1, \dots, s$: as Z is T -invariant, Z is also $(T-\lambda_i I)^{m_i}$ -invariant. In fact for any $\vec{x} \in Z$: $(T-\lambda_i I)^{m_i}(\vec{x})$

is a linear combination of vectors of the form $T^m(\vec{x})$ for $m=0, \dots, m_i$.

$$\text{As } T(\vec{x}) \in Z, \quad T^m(\vec{x}) \in Z \Rightarrow (T-\lambda_i I)^{m_i}(\vec{x}) \in Z.$$

$$\Rightarrow K_{\lambda_i} \cap Z = \text{null}(T-\lambda_i I)^{m_i} \quad \text{for each } i=1, \dots, s.$$

$$\text{Take } W = \text{Range}((T-\lambda_s I)^{m_s}|_Z) \subseteq Z.$$

Claim: 1) W is T -invariant

$$2) \quad (T-\lambda_s I)^{m_s}(K_{\lambda_i} \cap Z) = K_{\lambda_i} \cap Z \quad \text{and } K_{\lambda_i} \cap Z \subseteq W \quad \text{for all } i=1, 2, \dots, s-1$$

3) $\lambda_1, \dots, \lambda_{s-1}$ are eigenvalues of $T|_W$, and λ_s is not an eigenvalue of $T|_W$.

$$\text{Proof of Claim: 1) } \forall \vec{y} \in W: \exists \vec{x} \in Z \text{ s.t. } (T-\lambda_s I)^{m_s}(\vec{x}) = \vec{y}$$

$$\Rightarrow T(\vec{y}) = T((T-\lambda_s I)^{m_s}(\vec{x})) = (T-\lambda_s I)^{m_s}(T\vec{x}) \in W.$$

$$\Rightarrow W \text{ is } T\text{-invariant.}$$

$$2). \text{ For each } i=1, \dots, s: \quad \forall \vec{y} \in K_{\lambda_i} \cap Z: \exists p \in \mathbb{Z}^+ \text{ such that } (T-\lambda_i I)^p(\vec{y}) = \vec{0}$$

$$(T-\lambda_i)^p[(T-\lambda_s I)^{m_s}\vec{y}] = (T-\lambda_s I)^{m_s}(T-\lambda_i I)^p(\vec{y}) = \vec{0}$$

$$\Rightarrow (T-\lambda_s I)^{m_s}(\vec{y}) \in K_{\lambda_i}$$

$$\text{As } \vec{y} \in Z \quad \text{and } (T-\lambda_s I)^{m_s}(\vec{y}) \in Z$$

$$\Rightarrow K_{\lambda_i} \cap Z \text{ is } (T-\lambda_s I)^{m_s}\text{-invariant}$$

$$\text{Consider the restriction map: } (T-\lambda_s I)^{m_s}|_{K_{\lambda_i} \cap Z}: K_{\lambda_i} \cap Z \rightarrow K_{\lambda_i} \cap Z \quad \text{for each } i=1, \dots, s-1$$

$$\text{Since } \lambda_s \neq \lambda_i \text{ for any } i=1, \dots, s-1, \quad (T-\lambda_s I)^{m_s}|_{K_{\lambda_i} \cap Z} \text{ is one-to-one}$$

$$\text{Since } K_{\lambda_i} \cap Z \text{ is finite-dimensional, } (T-\lambda_s I)^{m_s}|_{K_{\lambda_i} \cap Z} \text{ is also onto.}$$

$$\Rightarrow (T-\lambda_s I)^{m_s}(K_{\lambda_i} \cap Z) = K_{\lambda_i} \cap Z.$$

3). Since $W \subseteq Z$, the only possible eigenvalues for $T|_W$ are $\lambda_1, \lambda_2, \dots, \lambda_s$.

As $\lambda_1, \dots, \lambda_s$ are eigenvalues for $T|_Z \Rightarrow E_{\lambda_i} \cap Z \neq \{\vec{0}\}$ for $i=1, \dots, s$

Since $\{\vec{0}\} \neq E_{\lambda_i} \cap Z \subseteq K_{\lambda_i} \cap Z \subseteq W$ ^{by part 2).} for all $i=1, 2, \dots, s-1$

$\Rightarrow \lambda_1, \dots, \lambda_{s-1}$ are eigenvalues of $T|_W$.

(To prove λ_s is not an eigenvalue of $T|_W$, we need to show if $\vec{y} \in W$ and $(T - \lambda_s I)(\vec{y}) = \vec{0}$, then $\vec{y} = \vec{0}$)

$\forall \vec{y} \in W$. $\exists \vec{x} \in Z$ such that $\vec{y} = (T - \lambda_s I)^{m_s}(\vec{x})$. Suppose $(T - \lambda_s I)(\vec{y}) = \vec{0}$.

$$\text{Then } \vec{0} = (T - \lambda_s I)(T - \lambda_s I)^{m_s}(\vec{x}) = (T - \lambda_s I)^{m_s+1}(\vec{x})$$

$$\Rightarrow \vec{x} \in K_{\lambda_s} \cap Z \subseteq \text{Null}((T - \lambda_s I)^{m_s})$$

$$\Rightarrow \vec{y} = (T - \lambda_s I)^{m_s}(\vec{x}) = \vec{0}.$$

$\Rightarrow \lambda_s$ is not an eigenvalue of $T|_W$. $\square$

(Note: W is a T -invariant subspace that we may apply our inductive hypothesis on).

$\forall \vec{x} \in Z$. $\vec{y} = (T - \lambda_s I)^{m_s}(\vec{x}) \in W$. By inductive hypothesis: $\exists \vec{y}_i \in K_{\lambda_i} \cap W$, for $i=1, \dots, s-1$

$$\vec{y} = \vec{y}_1 + \dots + \vec{y}_{s-1}. \quad (*)$$

As: $\vec{y}_i \in K_{\lambda_i} \cap W$. $\exists \vec{x}_i \in K_{\lambda_i} \cap Z$ such that $\vec{y}_i = (T - \lambda_i I)^{m_i}(\vec{x}_i)$

$$(*) \Leftrightarrow (T - \lambda_s I)^{m_s}(\vec{x}) = (T - \lambda_s I)^{m_s}(\vec{x}_1) + (T - \lambda_s I)^{m_s}(\vec{x}_2) + \dots + (T - \lambda_s I)^{m_s}(\vec{x}_{s-1})$$

$$\Leftrightarrow (T - \lambda_s I)^{m_s}(\vec{x} - \vec{x}_1 - \dots - \vec{x}_{s-1}) = \vec{0}$$

$$\Rightarrow \vec{x} - \vec{x}_1 - \dots - \vec{x}_{s-1} \in K_{\lambda_s} \cap Z$$

$$\Rightarrow \exists \vec{x}_s \in K_{\lambda_s} \cap Z, \text{ s.t. } \vec{x} - \vec{x}_1 - \dots - \vec{x}_{s-1} = \vec{x}_s$$

$$\Rightarrow \vec{x} = \vec{x}_1 + \dots + \vec{x}_s \text{ with } \vec{x}_i \in K_{\lambda_i} \cap Z. \quad \square$$

Theorem: Let $T: V \rightarrow V$ be a linear transformation. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be all the distinct eigenvalues of T on V with multiplicities $m_1, m_2, \dots, m_k$, respectively. Let $B_1, B_2, \dots, B_k$ be the bases for $K_{\lambda_1}, \dots, K_{\lambda_k}$, respectively.

- Then:
- 1). $B_i \cap B_j = \emptyset \quad \forall i \neq j$.
 - 2). $B = B_1 \cup B_2 \cup \dots \cup B_k$ forms a basis of V .
 - 3). $\dim K_{\lambda_i} = m_i$ for all $i=1, \dots, k$.

Proof: 1). $B_i \cap B_j \subseteq K_{\lambda_i} \cap K_{\lambda_j} = \{\vec{0}\}$ for any $\lambda_i \neq \lambda_j$

Since B_i and B_j are both linearly independent, $\vec{0} \notin B_i \cap B_j \Rightarrow B_i \cap B_j = \emptyset$.

$$3). |B_1 \cup B_2 \cup \dots \cup B_k| = \dim K_{\lambda_1} + \dots + \dim K_{\lambda_k} \leq m_1 + m_2 + \dots + m_k = n = \dim V.$$

Moreover, in the above inequality, the "=" is true if and only if $m_i = \dim K_{\lambda_i}$ for all $i=1, \dots, k$.

By the previous theorem, $\text{Span}(B_1 \cup B_2 \cup \dots \cup B_k) = V$

$$\Rightarrow |B_1 \cup B_2 \cup \dots \cup B_k| \geq \dim V.$$

We proved: $\dim V \leq |B_1 \cup B_2 \cup \dots \cup B_k| \leq \dim V \quad (*)$

$$\Rightarrow |B_1 \cup B_2 \cup \dots \cup B_k| = \dim V \quad \text{and} \quad \dim K_{\lambda_i} = m_i \quad \text{for all } i=1, \dots, k.$$

Since $\text{Span}(B_1 \cup \dots \cup B_k) = V$ and $|B_1 \cup B_2 \cup \dots \cup B_k| = \dim V$

$B_1 \cup B_2 \cup \dots \cup B_k$ is a basis of V .

Definition: Let $T: V \rightarrow V$ be a linear transformation. Let λ be an eigenvalue of T . Let $\vec{0} \neq \vec{x} \in K_{\lambda}$. Suppose

p is the smallest positive integer such that $(T - \lambda I)^p(\vec{x}) = \vec{0}$. Then

$\{(T - \lambda I)^{p-1}(\vec{x}), (T - \lambda I)^{p-2}(\vec{x}), (T - \lambda I)^{p-3}(\vec{x}), \dots, \vec{x}\}$ is called a cycle of generalized

eigenvectors associated with eigenvalue λ .

$(T - \lambda I)^{p-1}(\vec{x})$ is the initial vector of the cycle, $\vec{x}$ is the end vector of the cycle, and p is the length of the cycle.

Remarks: 1) The initial vector is an eigenvector associated with λ

2) The cycle contains linearly independent vectors.

3) Removing the end vector in a cycle with length ≥ 2 , the remaining set is still a cycle with the same initial vector.

4). Let $\gamma = \{(T-\lambda I)^{p_1}(\vec{x}), \dots, (T-\lambda I)(\vec{x}), \vec{x}\}$ be a cycle. Then $\text{Span } \gamma$ is T -invariant.

In fact. Let $y = a_1(T-\lambda I)^{p_1}(\vec{x}) + a_2(T-\lambda I)^{p_2}(\vec{x}) + \dots + a_p \vec{x} \in \text{Span } \gamma$. Then

$$(T-\lambda I)(\vec{y}) = a_2(T-\lambda I)^{p_1}(\vec{x}) + \dots + a_p(T-\lambda I)(\vec{x}) \Rightarrow T(\vec{y}) = a_2(T-\lambda I)^{p_1}(\vec{x}) + \dots + a_p(T-\lambda I)(\vec{x}) + \lambda \vec{y} \in \text{Span } \gamma.$$

5). Let $W = \text{Span } \gamma$ w/ $\gamma = \{(T-\lambda I)^{p_i}(\vec{x}), \dots, (T-\lambda I)(\vec{x}), \vec{x}\}$

$$\text{Denote: } \vec{y}_i = (T-\lambda I)^{p_i}(\vec{x}) \text{ for all } i=1, \dots, p$$

$$\text{i.e. } \gamma = \{ \overset{\text{I-th}}{\vec{y}_1}, \overset{\text{I-th}}{\vec{y}_2}, \dots, \vec{y}_p \}$$

$$\text{Then } (T-\lambda I)\vec{y}_i = \vec{y}_{i-1} \text{ for all } i=1, \dots, p$$

$$\Leftrightarrow T\vec{y}_i = \vec{y}_{i-1} + \lambda \vec{y}_i \Rightarrow [T\vec{y}_i]_{\gamma} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \lambda \\ 0 \\ \vdots \\ 0 \end{pmatrix} \begin{matrix} \rightarrow (i-1)\text{-th} \\ \rightarrow i\text{-th} \end{matrix}$$

$$\Rightarrow [T|_{\text{Span } \gamma}]_{\gamma} = \begin{pmatrix} \lambda & 1 & & \\ 0 & \lambda & & \\ \vdots & \vdots & \ddots & \\ 0 & 0 & & \lambda \end{pmatrix} \rightarrow \text{Jordan Block.}$$

6). If one can find a basis B of V consisting of cycles of generalized

eigenvectors of T . Then $[T]_B$ is in Jordan canonical form.

Lemma: Let $T: V \rightarrow V$ be a linear transformation. Let λ be an eigenvalue of T . Let $\gamma_1, \gamma_2, \dots, \gamma_q$

be cycles of T associated with λ such that initial vectors of $\gamma_1, \dots, \gamma_q$ are linearly independent. Then

(1). $\gamma_i \cap \gamma_j = \emptyset$ for all $i \neq j$, and

(2). $\bigcup_{i=1}^q \gamma_i$ is linearly independent.

Idea of Proof: By induction number of vectors in $\bigcup_{i=1}^q \gamma_i$, denote $n = |\bigcup_{i=1}^q \gamma_i|$.

Base case: When each cycle only contains exactly one vector. Then since each cycle only contains their initial vectors, by assumption, the cycles are linearly independent.

Inductive steps Suppose if $\gamma'_1, \dots, \gamma'_q$ are cycles such that $|\bigcup_{i=1}^q \gamma'_i| < n$ and the initial vectors of $\gamma'_1, \dots, \gamma'_q$ are linearly independent. Then $\bigcup_{i=1}^q \gamma'_i$ is linearly independent.

Then starting with $\gamma_1, \dots, \gamma_q$, remove the end vectors from each of $\gamma_1, \dots, \gamma_q$.

The resulting set are denoted by $\gamma'_1, \dots, \gamma'_q$. Then γ'_i is either $\emptyset$ or γ_i has $|\gamma_i| - 1$ vectors.

$\Rightarrow |\bigcup_{i=1}^q \gamma'_i| = n - q < n$. and the initial vectors of those cycles that are non-empty sets in $\gamma'_1, \dots, \gamma'_q$ is linearly independent

$\Rightarrow$ By inductive hypothesis $\bigcup_{i=1}^q \gamma'_i$ is linearly independent.

Now to prove $\bigcup_{i=1}^q \gamma_i$ is linearly independent. We set $W = \text{Span} \bigcup_{i=1}^q \gamma_i$ and show $\dim W = n$. To start with $\dim W \leq n$

Consider: $(T - \lambda I): W \rightarrow W$.

Then $\text{Range}(T - \lambda I) = \text{Span} \left(\bigcup_{i=1}^q \gamma'_i \right) \Rightarrow \dim \text{Range}(T - \lambda I) = n - q$

$\ker(T - \lambda I) = E_\lambda \cap W \Rightarrow \ker(T - \lambda I) \supseteq$ the set of initial vectors in $\gamma_1, \gamma_2, \dots, \gamma_q$

$\Rightarrow \dim(\ker(T - \lambda I)) \geq q$

$\Rightarrow n \geq \dim W \geq \dim \text{Range}(T - \lambda I|_W) + \dim(\ker(T - \lambda I)) \geq n - q + q = n$

$$\Rightarrow \dim W = n.$$

$$\Rightarrow \bigcup_{i=1}^r \mathcal{B}_i \text{ is linearly independent.}$$

Theorem: Let $T: V \rightarrow V$ be a linear transformation. Let λ be an eigenvalue of T . Then E_λ has a basis consisting of a union of disjoint cycles of generalized eigenvectors of λ .

Proof. Exercise.

Remark: The idea of constructing a basis of K_λ consisting of "disjoint cycles" is using a basis of E_λ .

In fact: Let $\{\vec{x}_1, \dots, \vec{x}_d\}$ be a basis of E_λ .

Then for each $\vec{x}_i$: Denote $\vec{y}_{i,1} = \vec{x}_i$, solve $\vec{y}_{i,2}$ s.t. $(T - \lambda I)\vec{y}_{i,2} = \vec{y}_{i,1}$
 $\vdots$
 $(T - \lambda I)\vec{y}_{i,k} = \vec{y}_{i,k-1}$

until the system: $(T - \lambda I)\vec{y}_{i,n_i+1} = \vec{y}_{i,n_i}$ is inconsistent.

Then $\{\vec{y}_{i,1}, \xrightarrow{T-\lambda I} \vec{y}_{i,2}, \xrightarrow{T-\lambda I} \vec{y}_{i,3}, \dots, \xrightarrow{T-\lambda I} \vec{y}_{i,n_i}\}$ is a cycle

The union of such cycles form a basis of K_λ .

Remarks: 1) # of Jordan blocks of $[T]_\mathcal{B}$ corresponding to $\lambda = \dim E_\lambda$.

2). $[T]_\mathcal{B}$ is diagonal iff and only if each Jordan block is 1×1 block

$$\Leftrightarrow \dim E_\lambda = m_\lambda.$$

end of March 10

Example: Find the Jordan canonical form of $A = \begin{pmatrix} 2 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -1 & 2 & 1 & 0 \\ 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$

Step 1 Find characteristic polynomial: $f(x) = \det(A - xI) = (x-2)^4$